The documentation and process conversion measures necessary to comply with this revision shall be completed by 27 Oct 1994.

INCH-POUND

MIL-S-19500/286E
27 July 1994
SUPERSEDING
MIL-S-19500/286D
15 March 1993

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER RECTIFIER,
MEDIUM-RECOVERY TYPES, 1N4245 THROUGH 1N4249
JAN, JANTX, JANTXV, AND JANHC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

These devices are no longer active for new design, preferred devices are - 1N5614, 1N5616, 1N5618, 1N5620, 1N5622 on MIL-S-19500/427

1. SCOPE

1.1 Scope. This specification covers the detail requirements for silicon, power rectifier, medium-recovery diodes. Three levels of product assurance are provided for each device type as specified in MIL-S-19500 and one level of product assurance for die.

1.2 Physical dimensions. See figure 1 and 2.

1.3 Maximum ratings.

Types	V_R	V_{RWM}	I_O 1/ T_A = 100°C 2/	I_{FSM} I_O = 1.0 A dc t_p = 8.3 ms
	V dc	V (pk)	A dc	A (pk)
1N4245	200	200	1.0	25
1N4246	400	400	1.0	25
1N4247	600	600	1.0	25
1N4248	800	800	1.0	25
1N4249	1000	1000	1.0	25

Types	Barometric pressure (reduced)	T_{STG} and T_J	$R_{\theta JL}$	t_{rr}
	mmHg	°C	°C/W	µs
1N4245	8	-65 to +175	42	5.0
1N4246	8	-65 to +175	42	5.0
1N4247	8	-65 to +175	42	5.0
1N4248	33	-65 to +175	42	5.0
1N4249	33	-65 to +175	42	5.0

1/ I_O rating is independent of heat sinking, special mounting, or forced air across the body or leads of the device.

2/ Derate linearly at 13.33 mA/°C between T_A = +100°C to +175°C.

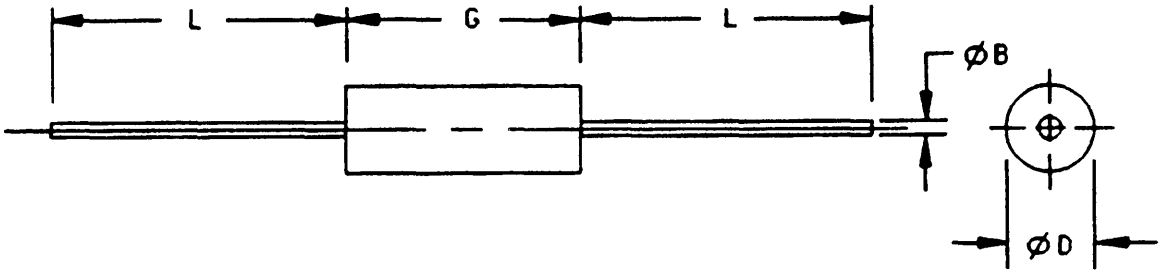
Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Research Laboratory, ATTN: AMSRL-EP-RD, Fort Monmouth, NJ 07703-5601 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5961

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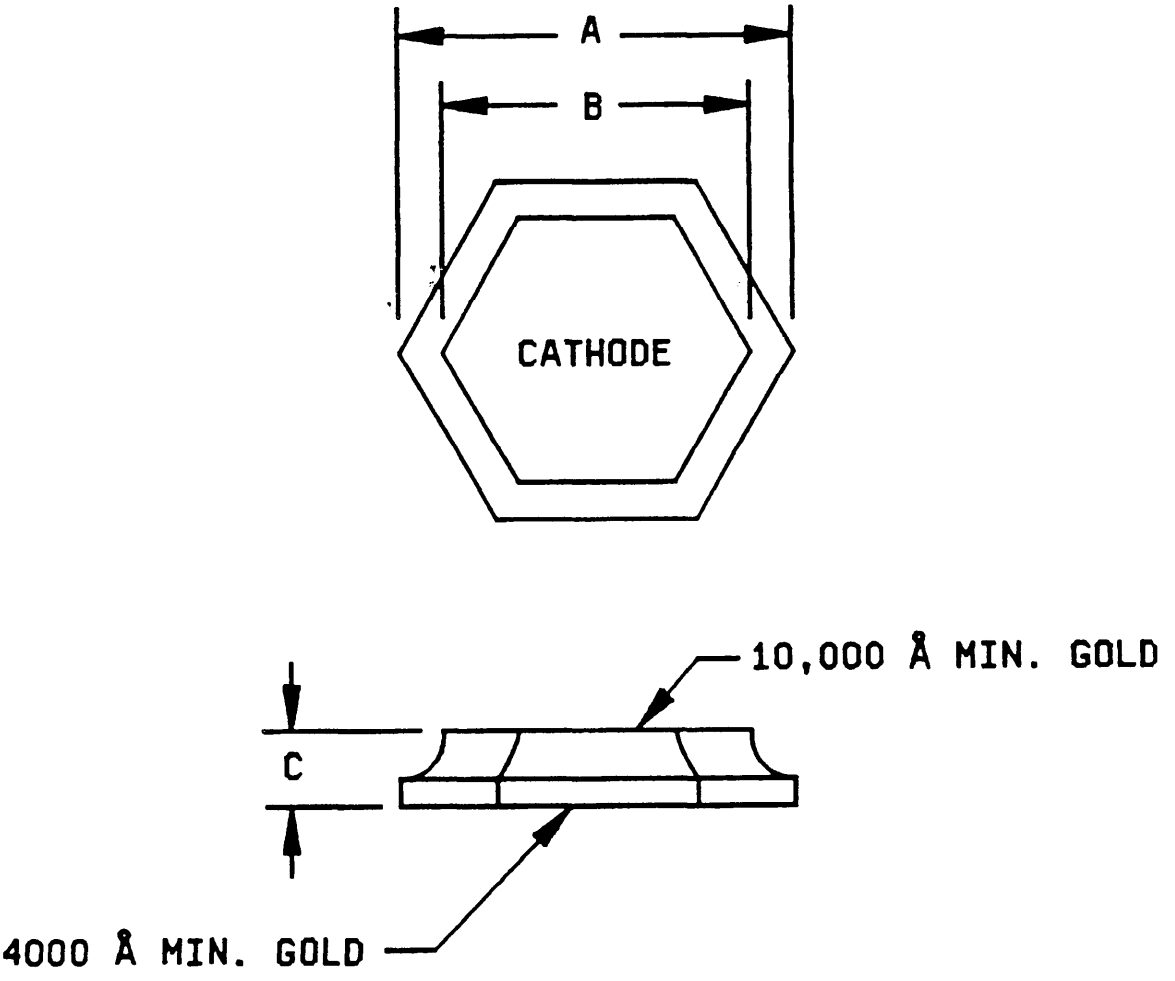


Dimensions					
Ltr.	Inches		Millimeters		Notes
	Min.	Max.	Min.	Max.	
ØB	.027	.033	0.69	0.84	
ØD	.065	.150	1.65	3.81	3
G	.140	.250	3.56	6.35	
L	1.000	1.500	25.40	38.10	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimension ØD shall be measured at the largest diameter.
4. The G dimension shall include all uncontrolled areas of the device leads.

FIGURE 1. Physical dimensions

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Symbol	Dimensions			
	Millimeters		Inches	
	Min	Max	Min	Max
A	1.19	1.35	.047	.053
B	0.84	0.94	.033	.037
C	0.18	0.28	.007	.011

FIGURE 2. JANHC (A-version) die dimensions.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500 and on figure 1 and 2 herein. No lead (Pb) shall be used in the construction of the die bonds.

3.3.1 Lead material and finish. Lead material shall be solderable as defined in MIL-S-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.3.2 Diode construction. Except for JANHC these devices shall be constructed utilizing metallurgically bonded, thermally matched non-cavity, double plug construction. The metallurgical bond between the chip and the plug shall meet the category I requirement of MIL-S-19500, appendix A.

3.4 Marking. Marking shall be in accordance with MIL-S-19500.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

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4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

4.2.1 JANC devices. Qualification for JANC devices shall be in accordance with appendix H of MIL-S-19500.

4.3 Screening (JANTX and JANTXV levels only). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement
	JANTX and JANTXV
9	Not applicable
11	I_{R1} and V_{F1}
12	See 4.3.1
13	Subgroup 2 of table I herein. $\Delta V_{F1} \leq +.1, -.2$ V dc $\Delta I_{R1} \leq$ 100 percent of initial value or ±100 nA dc, whichever is greater. Scope display evaluation (see 4.5.2)

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows (see 4.5.1):

I_O = 1 amp minimum, $V_R = V_{RWM}$, T_A = room ambient as defined in the general requirements of (see 4.5) of MIL-STD-750; f = 50-60 Hz.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVb (JANTX, and JANTXV) of MIL-S-19500, and as follows. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, group A, subgroup 2 herein.

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4.4.2.1 Group B inspection, table IVb of MIL-S-19500.

Subgroup	Method	Condition
2	4066	I_{FSM} = 25 A (pk), 10 surges of 1/120 s each at 1 minute intervals, super-imposed on I_O = 1.0 A dc, V_R = rated V_{RWM} (see 1.3)
3	1027	I_O = 1.0 A dc minimum; T_A = +100°C minimum; T_J = +145°C minimum; f = 50-60 Hz; V_R = rated V_{RWM} (see 1.3 and 4.5.1). NOTE: I_O and/or T_A may be increased to achieve a T_J = +145°C minimum.
5	3101	+25°C ≤ T_R +35°C , $R_{\theta JL}$ = 42 °C/W, L = 3/8".

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and as follows. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, group A, subgroup 2 herein.

4.4.3.1 Group C inspection, table V of MIL-S-19500.

Subgroup	Method	Condition
2	2036	Terminal strength (lead fatigue), test condition E.
5	1001	5 devices, c = 0, pressure = 8 mmHg for 1N4245, 1N4246, and 1N4247; 33 mmHg for 1N4248 and 1N4249.
6	1026	I_O = 1.0 A dc minimum; T_A = +100°C minimum; T_J = +145°C minimum; f = 50-60 Hz; V_R = rated V_{RWM} (see 1.3 and 4.5.1). NOTE: I_O and/or T_A may be increased to achieve a T_J = +145°C minimum.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables as follows.

4.5.1 Burn-in and steady-state operation life tests. These tests shall be conducted with a half-sine waveform of the specified peak voltage impressed across the diode in the reverse direction followed by a half-sine waveform of the specified average rectified current. The forward conduction angle of the rectified current shall be neither greater than 180°, nor less than 150°.

4.5.2 Scope display evaluation. The reverse breakdown characteristics shall be viewed on an oscilloscope with display calibration factors of 5 to 20 μA/division and 50 to 200 V/division. Reverse current over the knee shall be at least 50 μA. Each device may exhibit a sharp knee characteristic and any discontinuity or dynamic instability of the trace shall be cause for rejection.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Forward voltage	4011	I_F = 3.0 A (pk), duty cycle ≤ 2% pulsed, tp ≤ 8.3 ms	V_{F1}	0.6	1.3	V (pk)
Reverse current leakage	4016	DC method; V_R = rated (see 1.3)	I_{R1}		1.0	µA dc
Breakdown voltage	4021	I_R = 100 µA dc	$V_{(BR)}$			
1N4245				240		V
1N4246				480		V
1N4247				720		V
1N4248				960		V
1N4249				1150		V
Subgroup 3						
High temperature operation:		T_A = +150°C				
Reverse current leakage	4016	DC method, V_R = rated (see 1.3)	I_{R2}		150	µA dc
Low temperature operation:		T_A = -65°C				
Forward voltage	4011	I_F = 3.0 A (pk); duty cycle ≤ 2% pulsed; tp ≤ 8.3 ms	V_{F2}		1.5	V dc
Subgroup 4						
Reverse recovery time	4031	Condition B, I_F = 0.5 A, I_R = 1.0 A, $I_{(REC)}$ = 0.25 A	t_{rr}		5	µs

1/ For sampling plan, see MIL-S-19500.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish as specified (see 3.3.1).

c. Product assurance level and type designation.

6.3 Suppliers of die. The qualified die suppliers with the applicable letter version (example JANHCA1N4461) will be identified on the QPL.

JANC ordering information		
	Manufacturer	
PIN	14552	
1N4245	JANHCA4245	
1N4246	JANHCA4246	
1N4247	JANHCA4247	
1N4248	JANHCA4248	
1N4249	JANHCA4249	

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - AS, CG, MC
Air Force - 15, 19, 85, 99
DLA - ES

Preparing activity:
Army - ER

Agent:
DLA - ES

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1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/286E	2. DOCUMENT DATE (YYMMDD)

3. DOCUMENT TITLE

SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER RECTIFIER, MEDIUM-RECOVERY, TYPES 1N4245 THROUGH 1N4249 JAN, JANTX, JANTXV, AND JANHC

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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